Experiment No. 2

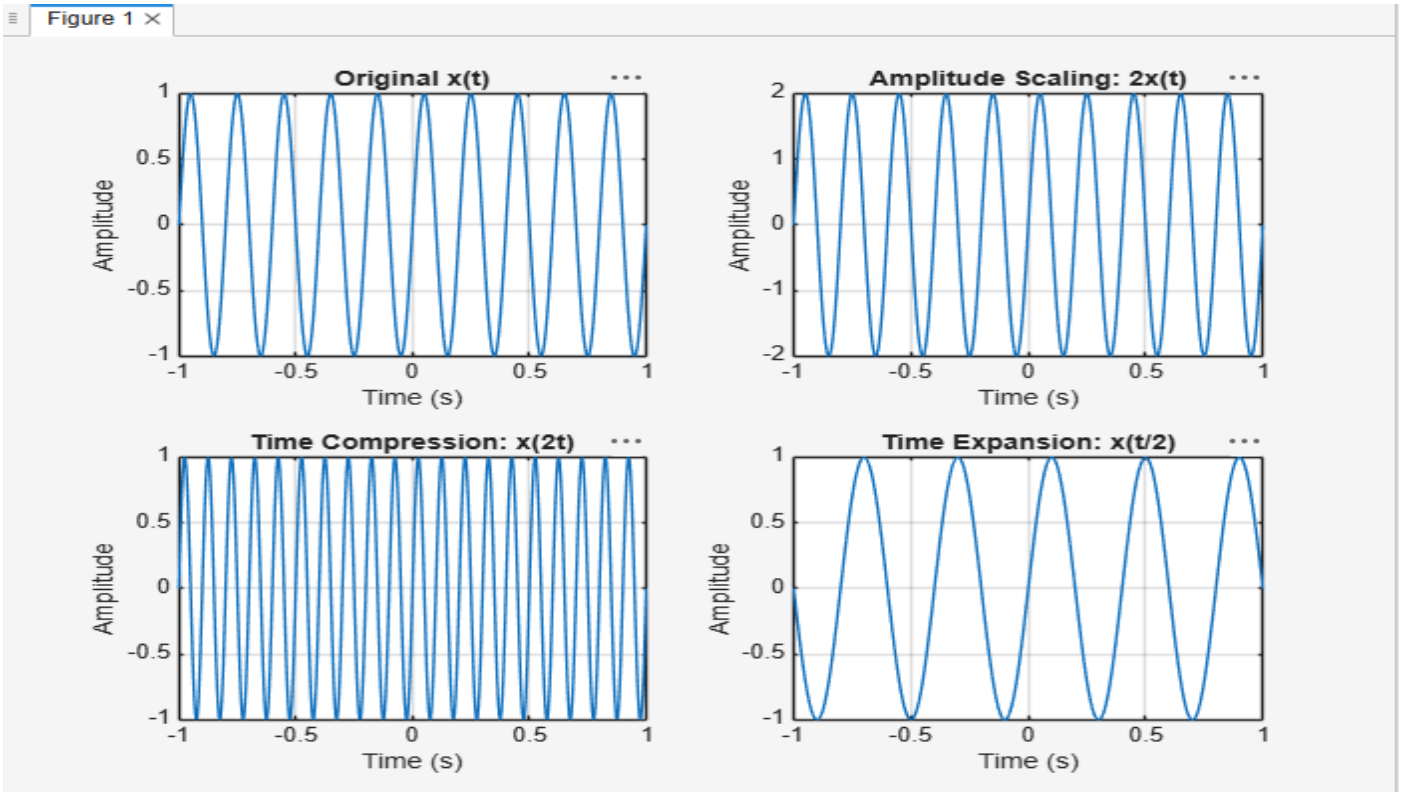
Signal Operations and Transformations

(i)

/MATLAB Drive/Experiment_2_1.m

```
1   clc; clear; close all;
2   t = -1:0.001:1;
3   x = sin(2*pi*5*t);
4   yA = 2*x;
5   yC = sin(2*pi*5*(2*t));
6   yE = sin(2*pi*5*(t/2));
7   figure;
8   subplot(2,2,1); plot(t,x); grid on;
9   title('Original x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
10  subplot(2,2,2); plot(t,yA); grid on;
11  title('Amplitude Scaling: 2x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
12  subplot(2,2,3); plot(t,yC); grid on;
13  title('Time Compression: x(2t)'); xlabel('Time (s)'); ylabel('Amplitude');
14  subplot(2,2,4); plot(t,yE); grid on;
15  title('Time Expansion: x(t/2)'); xlabel('Time (s)'); ylabel('Amplitude');
```

Code



Output